

Portable EEG Correlates of Basketball Free-Throw Performance: Multi-Block Session Analysis Report

Lukas Laskowski

Kyle E. Mathewson

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Abstract

This report presents the analysis of a multi-block free-throw session recorded using the FreethrowEEG system. The participant performed 3 blocks of 40 free throws each, yielding 78 analysed shots (34 made, 44 missed; 44%) after data quality screening and exclusion of experimenter-flagged trials. Continuous EEG band power (delta, theta, alpha, beta, gamma) was recorded from a Muse 2 headset. Shot-locked epochs were extracted and compared between successful and unsuccessful attempts. Video was recorded for two of the three blocks; post-hoc pose estimation using MediaPipe was performed on the available video to extract biomechanical features aligned with neural dynamics. Descriptive statistics, time-series visualizations, pose kinematics, and exploratory inferential tests are reported. Data were collected across 3 blocks of 40 shots each. After data quality screening and exclusion of experimenter-flagged trials, 78 shots were retained for EEG analysis.

1 Introduction

Electroencephalography (EEG) has been widely used to study neural correlates of motor performance, including precision sport tasks such as archery, shooting, and golf putting. A consistent finding in the literature is that successful performers exhibit distinct pre-performance EEG patterns, particularly reduced alpha desynchronization and modulated theta activity, compared to unsuccessful attempts. This session tests whether such patterns are detectable using a portable, four-channel Muse 2 headset during basketball free-throw shooting.

This report extends an initial pilot by using real EEG data collected over multiple blocks, providing substantially more statistical power ($N = 78$ shots) and improved experimental control including better video framing and consistent shot-cue logging.

The present analysis focuses on:

1. Visualizing continuous and shot-locked EEG band power;
2. Comparing pre-shot, execution, and post-shot neural signatures between made and missed free throws;
3. Evaluating the theta/alpha ratio as a candidate readiness marker;
4. Examining shot-by-shot band power progression across the session;
5. Biomechanical pose analysis from concurrent video (where available).

2 Methods

2.1 Participant and Task

The participant (Lukas) performed 3 blocks of 40 free throws (120 total), collected in a public gymnasium on a single day. The full session spanned approximately 51.8 minutes of active recording. Each trial followed a structured protocol: a preparation phase (~ 5 s), a pre-shot baseline (~ 5 s), shot execution (~ 2 s), post-shot recording (~ 3 s), and a review phase. Shot outcome (made/missed) was coded by the participant.

2.2 Data Quality and Exclusions

Block 1: 3 analysed of 40 shots (0 excluded, 37 invalid EEG; with video); Block 2: 39 analysed of 40 shots (1 excluded, 0 invalid EEG; no video); Block 3: 36 analysed of 40 shots (4 excluded, 0 invalid EEG; with video). In Block 1, an EEG recording buffer issue caused the raw data stream to stop after approximately 3 minutes, yielding valid EEG for only the first 3 shots; the remaining 37 shots in that block had degenerate timestamps and were excluded from analysis. The experimenter also reported missed shot cues on certain trials: Block 1 shots 18 and 35, Block 2 shot 34, and Block 3 shots 8, 21, 22, and 25 were excluded. After all filtering, 78 shots were retained for EEG analysis (34 made, 44 missed; 44% accuracy).

Video was recorded for Blocks 1 and 3 but not for Block 2 (the video file did not save). Due to the EEG timing issues in Block 1, video analysis was performed primarily on Block 3 (36 shots).

2.3 EEG Recording

EEG was recorded continuously using a Muse 2 headband (four channels: TP9, AF7, AF8, TP10; 256 Hz sampling rate). Band power was computed in real time using a 1-second sliding window with 25% overlap. A 4th-order Butterworth bandpass filter was applied to extract five standard frequency bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–50 Hz). Environmental line noise was removed via a Butterworth notch filter at 60 Hz. For the continuous power analysis, raw electrode signals were averaged across all four channels and band power was recomputed offline.

2.4 Video Recording and Pose Estimation

Video was recorded simultaneously at 30 fps (WebM/VP9 codec) using the device camera. Post-hoc pose estimation was performed using the MediaPipe Pose Landmarker (heavy model), extracting 33 body landmarks per frame. Biomechanical features—elbow angle (shoulder–elbow–wrist), wrist height, knee angle (hip–knee–ankle), and body lean angle (torso deviation from vertical)—were computed for each frame during the shot execution phase. Ball release was estimated as the frame where wrist velocity peaked while the wrist was near its highest position.

2.5 Analysis

Continuous EEG data were segmented into shot-locked epochs aligned to the onset of the execution phase. Epochs were interpolated onto a common time grid (-12 to $+8$ s relative to shot onset at 0.25 s resolution) and averaged separately for made and missed shots. Phase-wise mean power was compared using Welch’s t -test (unequal variances) and the Mann–Whitney U test. Effect sizes are reported as Cohen’s d .

All 78 shots with valid EEG data across all three blocks were used for the EEG analyses (Figures 1–8 and Table 1). Video-based analyses (Figures 9–13) used only the subset of shots from blocks with available video.

Individual shot clips were extracted from the video, and stop-motion frame sequences were generated for each trial. Average frame composites were created by pixel-averaging release-point frames within each outcome group. “Ghost” overlays show the variability in shooting form by superimposing skeleton renderings from all shots within a group.

3 Results

3.1 Session Overview

The participant completed 78 analysed shots across 3 blocks over approximately 51.8 minutes of active recording (44% accuracy). Figure 1 shows the session timeline with shot phases and outcomes overlaid on the continuous alpha power trace. Block boundaries are indicated by vertical dashed lines.

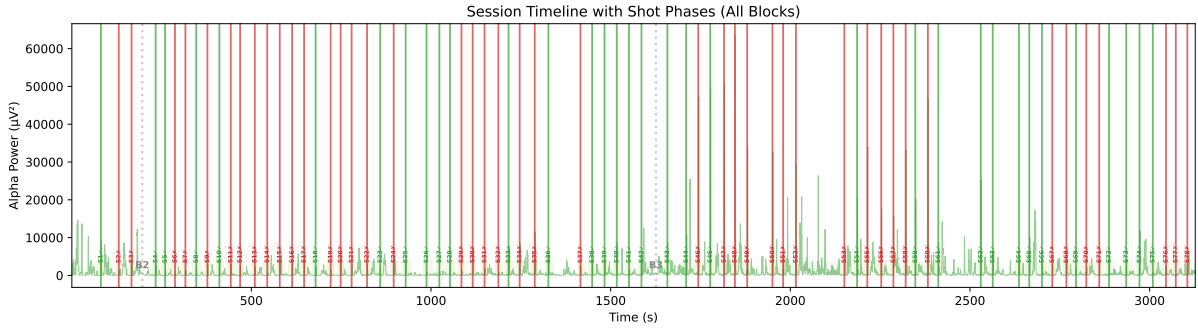


Figure 1: Session timeline. Each vertical line marks a shot (green = made, red = missed). Shaded regions indicate trial phases. Dotted grey lines mark block boundaries.

3.2 Continuous Band Power

Figure 2 shows the five frequency bands across all blocks. Visual inspection of the continuous traces reveals characteristic fluctuations across all five bands over the 51.8-minute session.

Continuous Band Power Over Session (All Blocks)



Figure 2: Continuous band power (delta through gamma) over all blocks. Band power was computed offline from the average of four electrode channels. Dashed vertical lines indicate shot execution (green = made, red = missed).

3.3 Signal Filtering

Figure 3 demonstrates the effect of low-pass, high-pass, and bandpass Butterworth filtering applied to the raw delta-band signal from the first 60 seconds of recording.

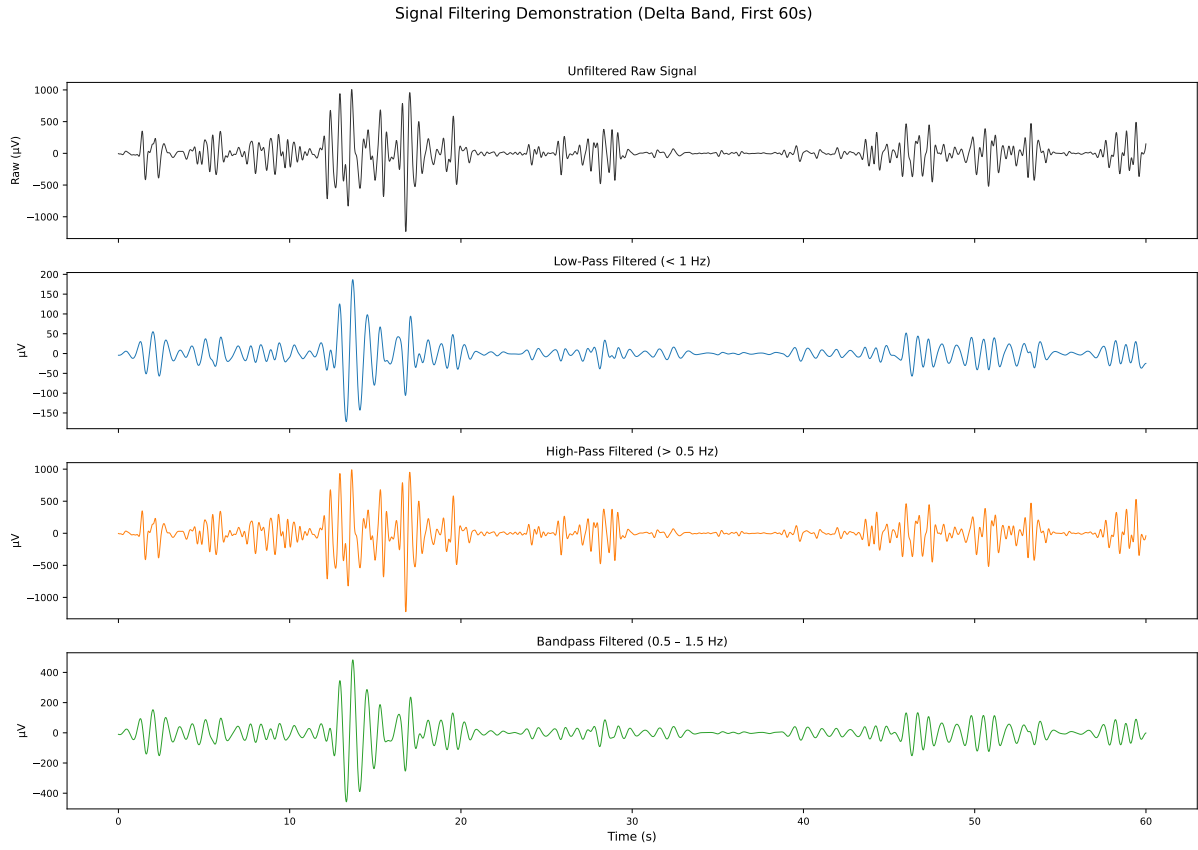


Figure 3: Filtering demonstration on the raw delta signal: (a) unfiltered, (b) low-pass <1 Hz, (c) high-pass >0.5 Hz, (d) bandpass 0.5–1.5 Hz.

3.4 Shot-Locked Averages

Figure 4 shows mean band power ($\pm$ SEM) locked to the onset of the execution phase, averaged across all 78 shots from all blocks.

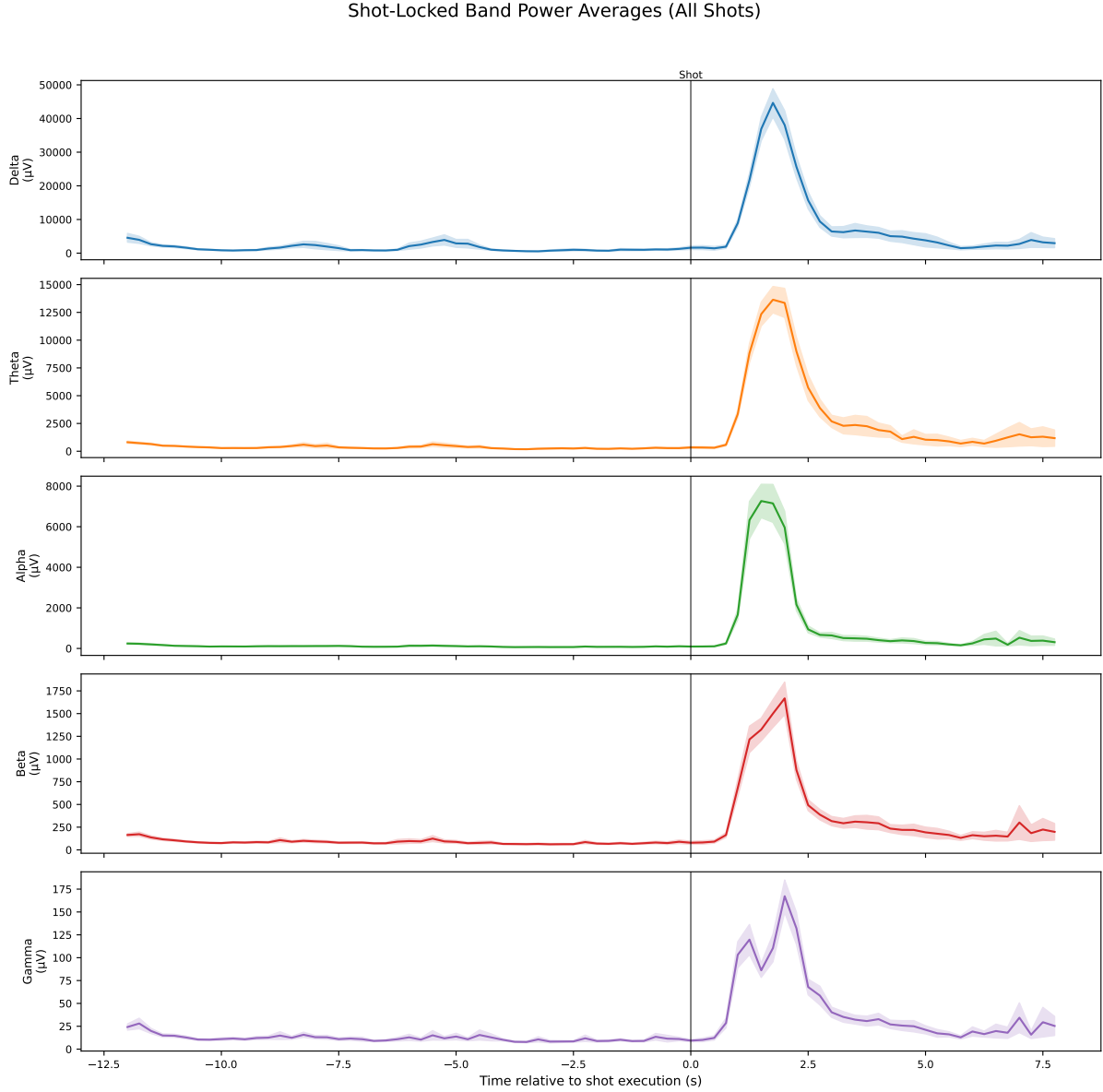


Figure 4: Shot-locked band power averages (all shots). Time 0 = execution onset. Shaded regions represent ± 1 SEM.

3.5 Made vs. Missed Comparison

Figure 5 compares shot-locked averages for made and missed shots. delta power was 50.9% higher for made shots ($M = 1717.7$) compared to missed ($M = 1138.0$). theta power was 39.6% higher for made shots ($M = 390.4$) compared to missed ($M = 279.5$). alpha power was 16.7% higher for made shots ($M = 105.9$) compared to missed ($M = 90.8$). beta power was 9.7% higher for made shots ($M = 84.7$) compared to missed ($M = 77.2$). gamma power was 6.9% higher for made shots ($M = 11.6$) compared to missed ($M = 10.9$). No frequency band reached trend-level significance ($p < .10$) in the pre-shot comparison, consistent with the limited sample size.

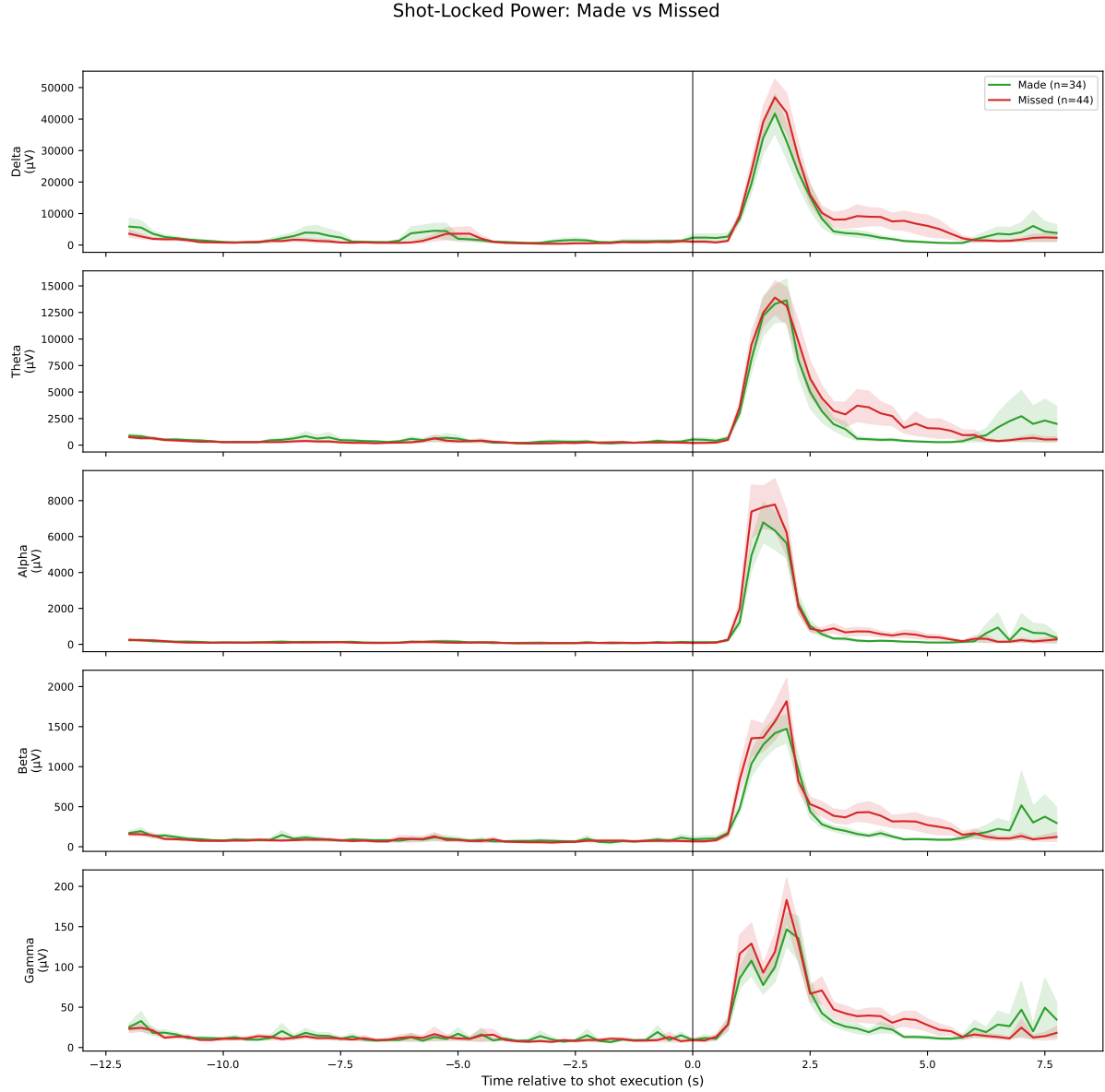


Figure 5: Shot-locked power comparison: made (green) vs. missed (red). Shaded regions are ± 1 SEM.

3.6 Phase-Wise Analysis

Figure 6 displays mean power during the pre-shot, execution, and post-shot phases, broken down by outcome and frequency band. For made shots, delta power increased by 828.5% from pre-shot to execution. For made shots, theta power increased by 1358.7% from pre-shot to execution. For made shots, alpha power increased by 2205.9% from pre-shot to execution. For made shots, beta power increased by 667.9% from pre-shot to execution. For made shots, gamma power increased by 491.2% from pre-shot to execution.

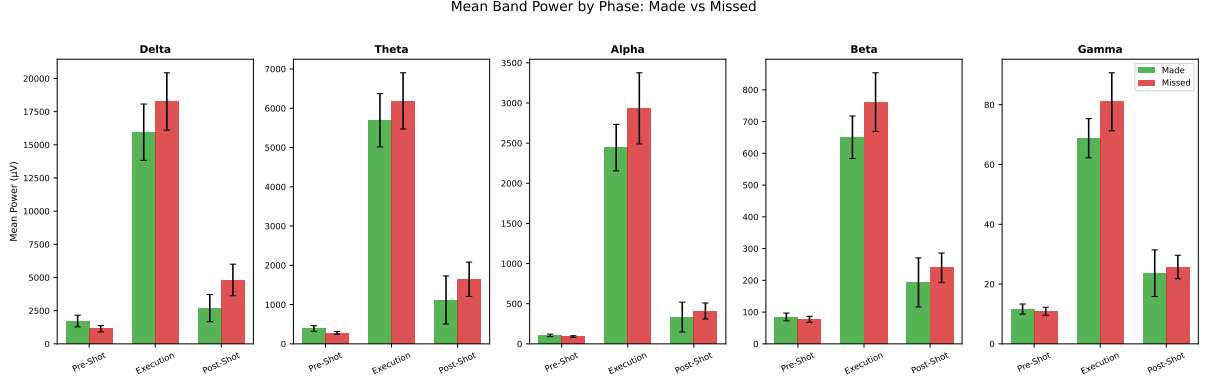


Figure 6: Mean band power by phase and outcome ($\pm$ SEM).

3.7 Statistical Comparisons

Table 1 summarizes pre-shot band power for made and missed shots, along with inferential test results.

Table 1: Pre-shot band power: descriptive and inferential statistics.

Band	Made M (SD)	Missed M (SD)	t	p	U	d
Delta	1717.7 (2600.6)	1138.0 (1577.9)	1.15	0.257	846	0.28
Theta	390.4 (426.2)	279.5 (243.1)	1.36	0.182	818	0.33
Alpha	105.9 (89.3)	90.8 (69.5)	0.82	0.416	789	0.19
Beta	84.7 (72.3)	77.2 (63.2)	0.48	0.633	766	0.11
Gamma	11.6 (10.0)	10.9 (9.0)	0.34	0.733	748	0.08

3.8 Theta/Alpha Ratio

The pre-shot theta/alpha ratio has been proposed as an index of focused attention in sport performance.. Mean pre-shot θ/α ratio was 3.53 (SD = 1.25) for made shots and 3.24 (SD = 0.98) for missed shots. Made shots showed a higher θ/α ratio, differing by 0.29.

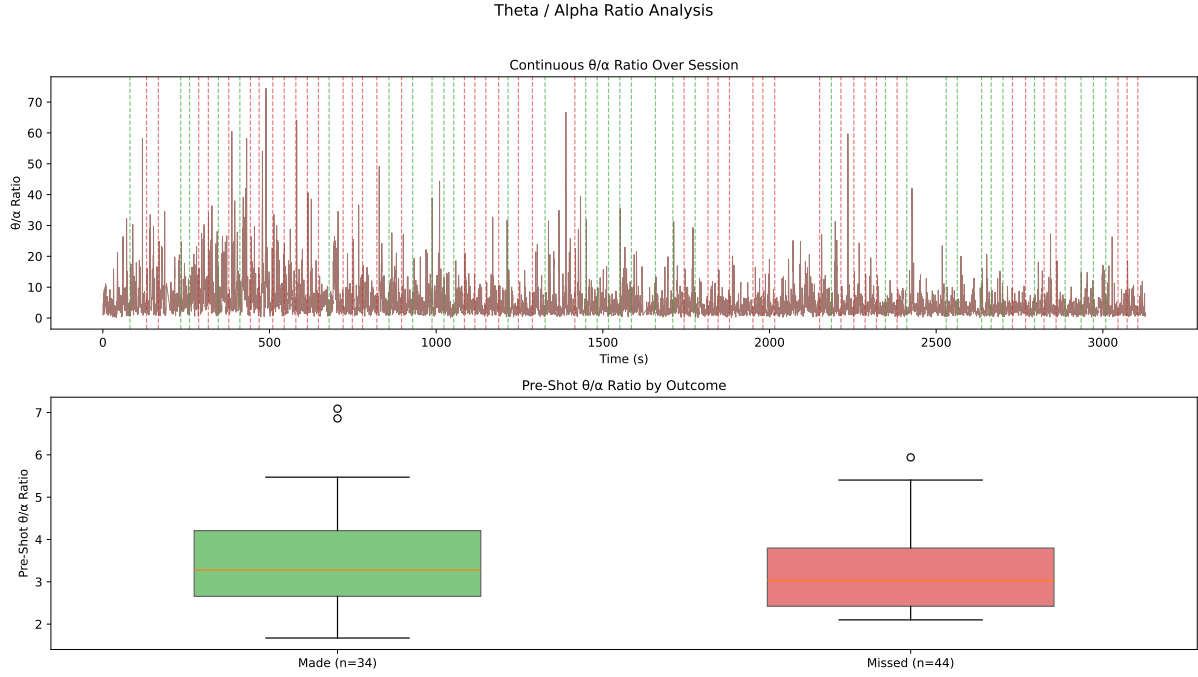


Figure 7: (Top) Continuous θ/α ratio over the session. (Bottom) Pre-shot θ/α ratio by outcome.

3.9 Shot-by-Shot Progression

Figure 8 tracks pre-shot band power across the 78 shots. Pre-shot alpha power showed a trend toward increase across the session ($r = 0.31$, $p = 0.006$), though the correlation was weak, possibly due to the limited number of trials.

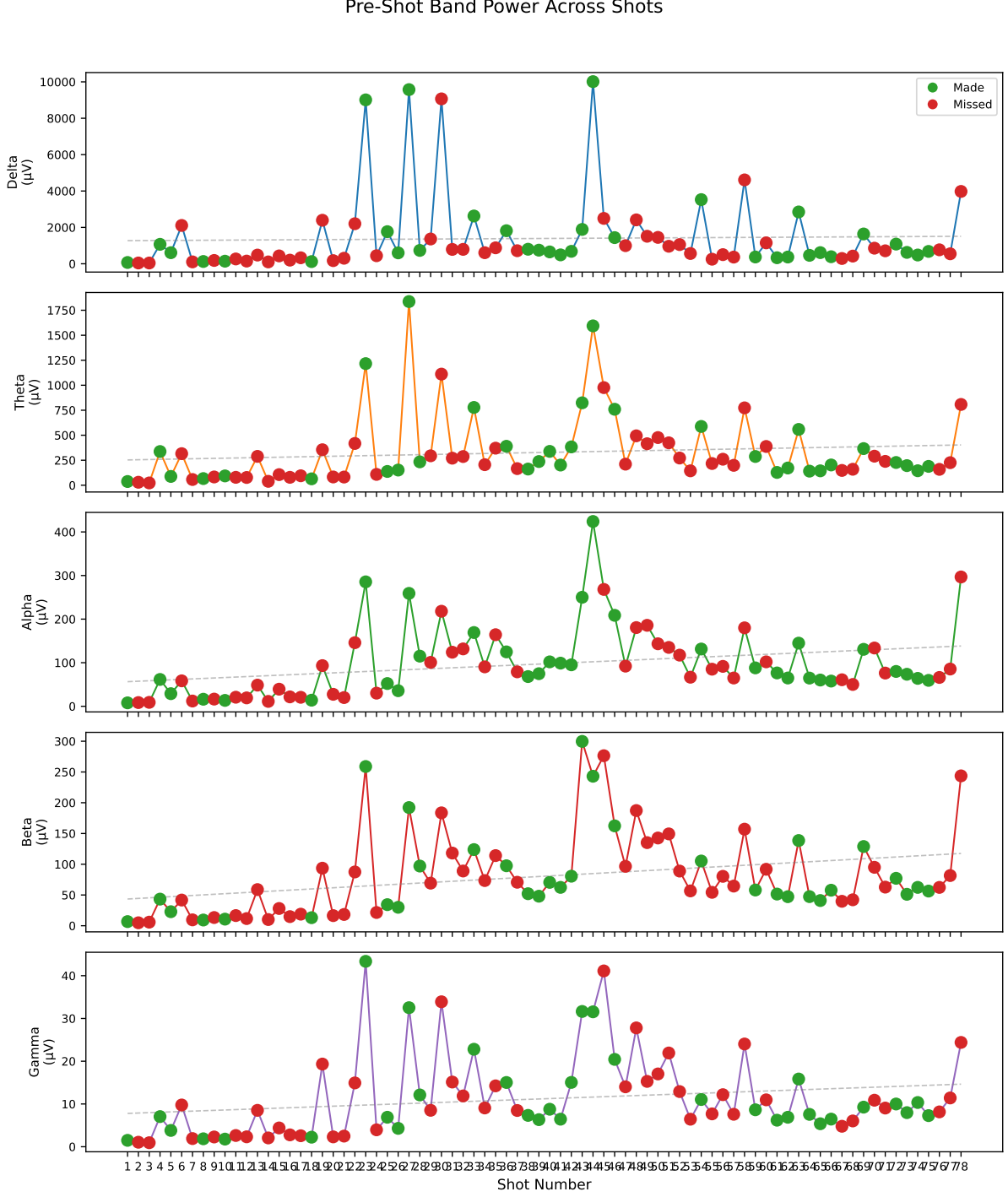


Figure 8: Pre-shot band power across shots. Dot colour indicates outcome (green = made, red = missed). Dashed line shows linear trend.

4 Video and Pose Estimation Analysis

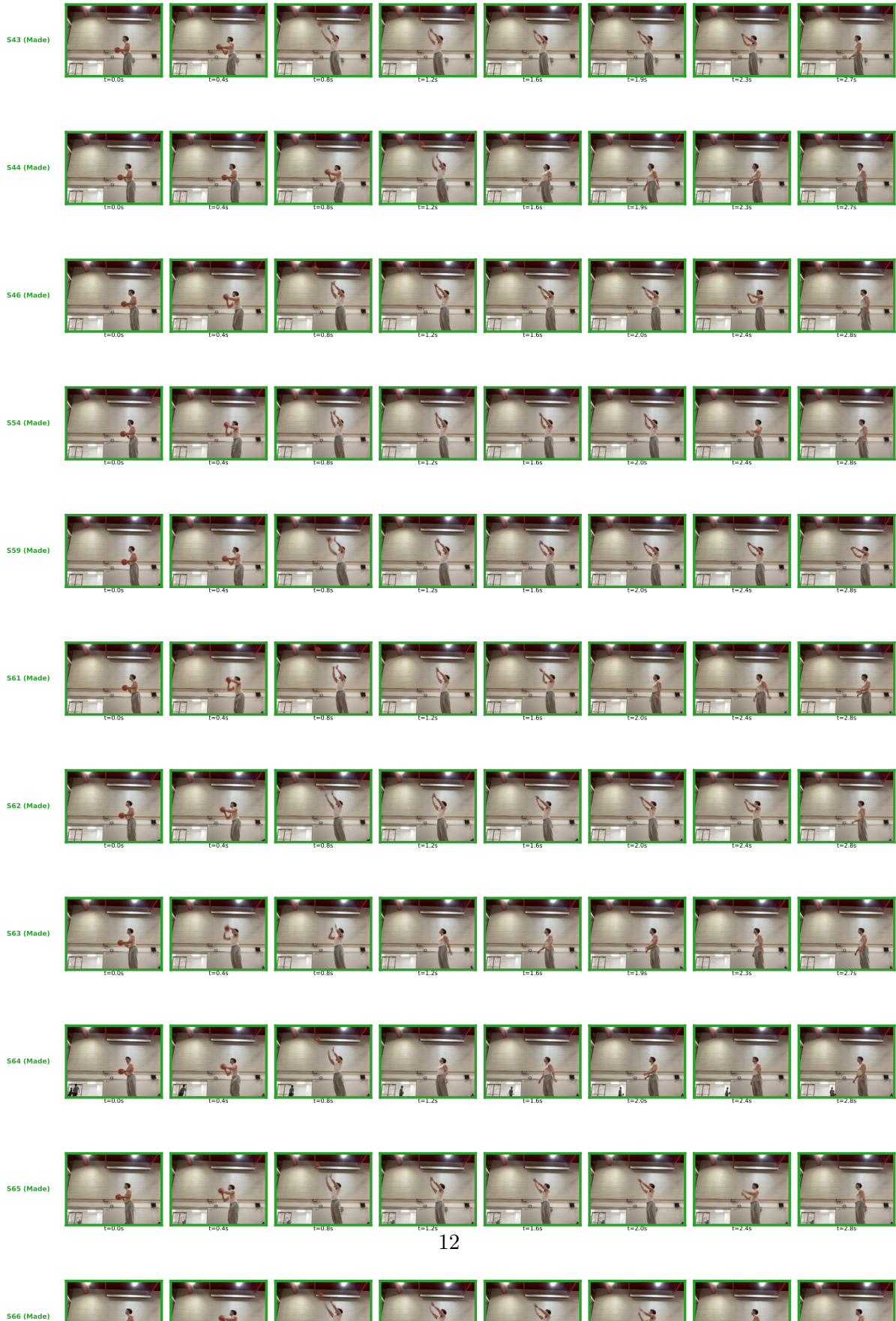
Video was recorded for Blocks 1 and 3, but not for Block 2 (video did not save). Due to EEG timing synchronization issues in Block 1, video-based analyses were performed on Block 3's 36 shots only. Post-hoc pose estimation was performed using MediaPipe Pose (heavy model) to extract 33 body landmarks per frame. Biomechanical features—elbow angle, wrist height, knee

angle, and body lean angle—were computed across the shot execution phase for each trial and compared between made and missed shots.

4.1 Shot Stop-Motion Montage

Figure 9 presents a frame-by-frame stop-motion decomposition of each shot during the execution phase, with pose skeleton overlays. Made shots are shown in the upper rows (green borders) and missed shots in the lower rows (red borders).

Figure 9 — Shot Stop-Motion Montage



4.2 Ghost Shot Overlay

Figure 10 shows composite “ghost” images created by pixel-averaging the release-point frames across made and missed shots separately (top row), along with skeleton-only overlays showing individual shot pose variability (bottom row).

Figure 10 — Ghost Shot Overlay at Release

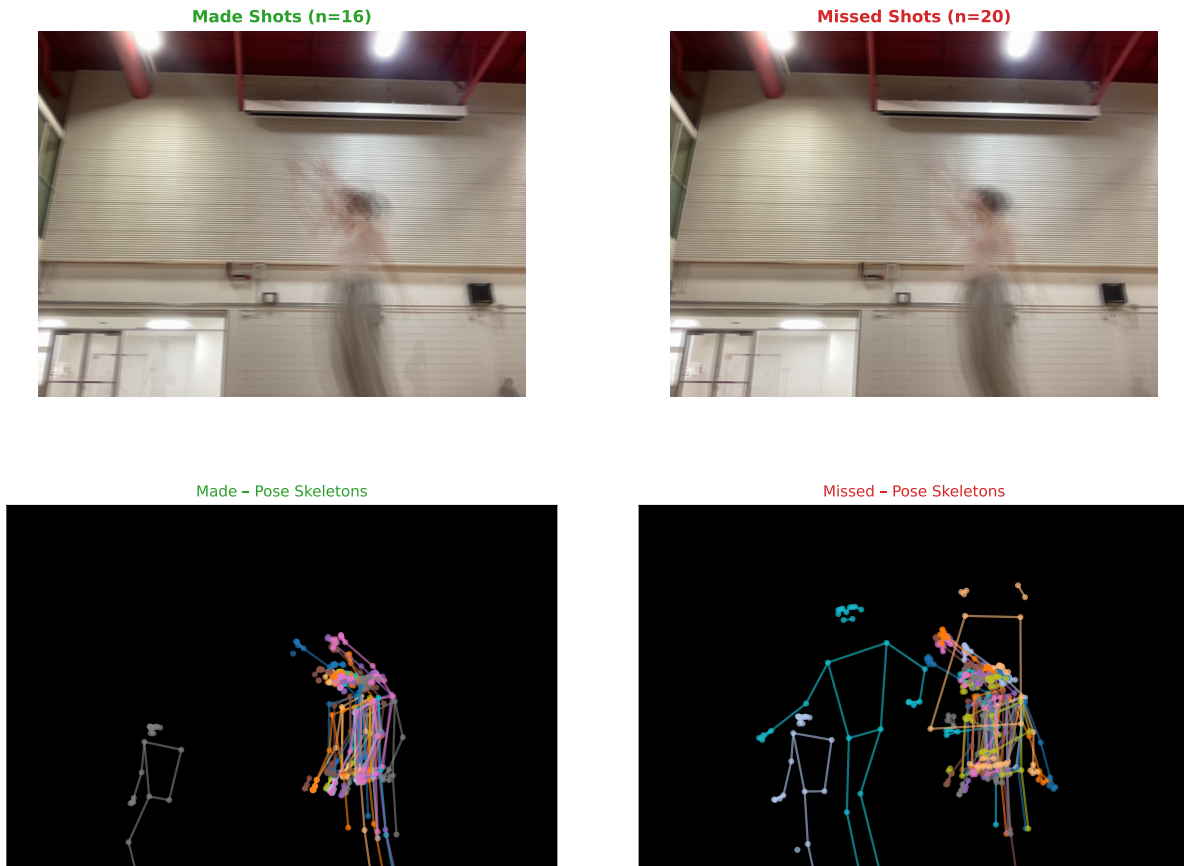


Figure 10: Ghost overlay composites at the estimated release point. Top: pixel-averaged frames for made (left) and missed (right) shots. Bottom: skeleton-only overlays with individual shots in distinct colours.

4.3 Pose Kinematics Trajectories

Figure 11 plots the time course of four biomechanical features during the execution phase, comparing made (green) and missed (red) shots. Peak elbow angle was lower for made shots (170.5) than missed (175.2). Peak wrist height was lower for made shots (0.7) than missed (0.8). Peak knee angle was lower for made shots (176.0) than missed (177.1). Peak body lean was lower for made shots (18.7) than missed (31.5).

Figure 11 — Pose Kinematics During Shot Execution (Release-Aligned)

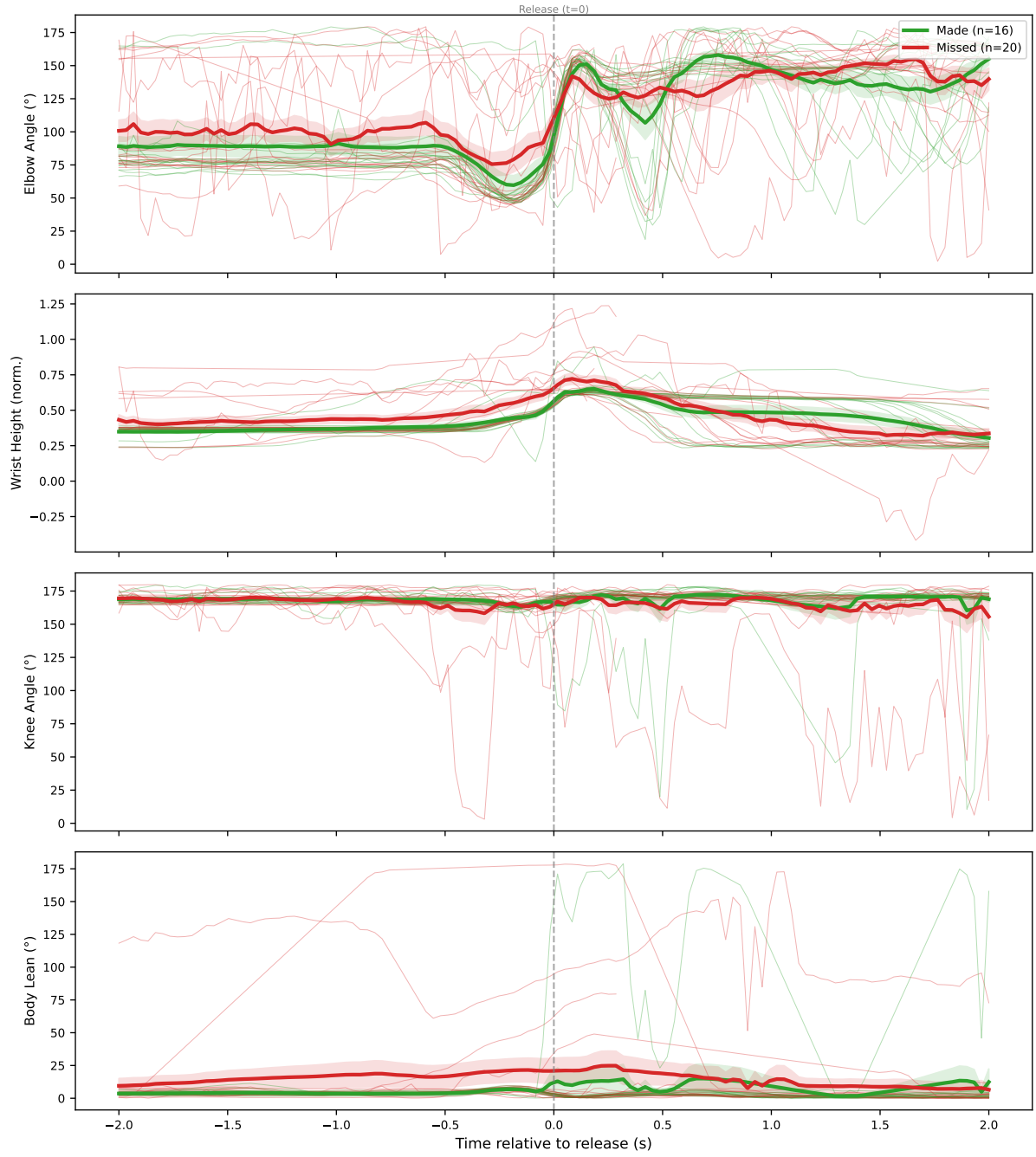


Figure 11: Biomechanical feature trajectories during shot execution (mean $\pm$ SEM). Thin lines show individual shots; thick lines show group means. Dashed vertical line: estimated release point.

4.4 Integrated Pose and EEG Analysis

Figure 12 presents a joint visualization of pose kinematics (elbow angle, wrist height) and EEG dynamics (alpha power, θ/α ratio) during the execution phase, time-aligned to recording onset. Time-aligned visualization of pose and EEG data during execution reveals concurrent motor and neural dynamics. Cross-domain correlations between mean pose features and alpha power

are annotated where available.

Figure 12 — Pose Kinematics + EEG During Execution

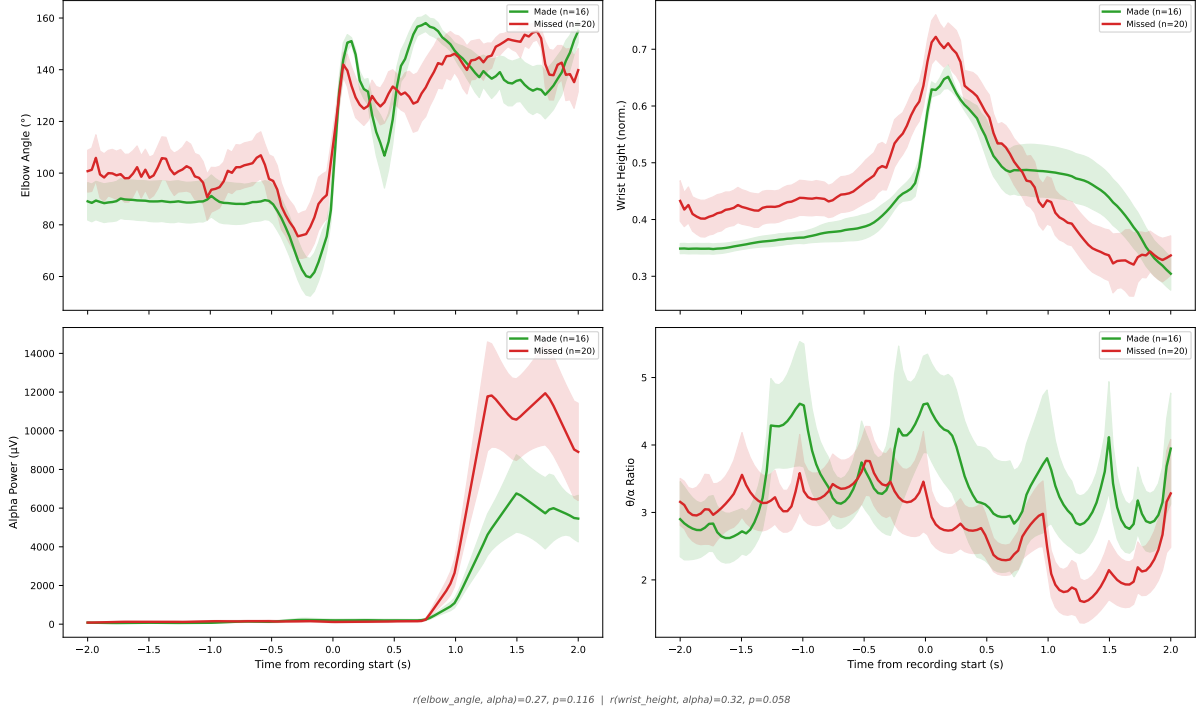


Figure 12: Combined pose kinematics and EEG dynamics during shot execution. Top row: elbow angle and wrist height. Bottom row: alpha band power and θ/α ratio. All traces show made (green) vs. missed (red) mean $\pm$ SEM.

4.5 Average Frame Composites

Figure 13 shows pixel-averaged video frames at three time points (start, midpoint, end) of the execution phase for made and missed shots, with an absolute difference image highlighting regions of greatest divergence.

Figure 13 — Average Frame Composites: Made vs Missed

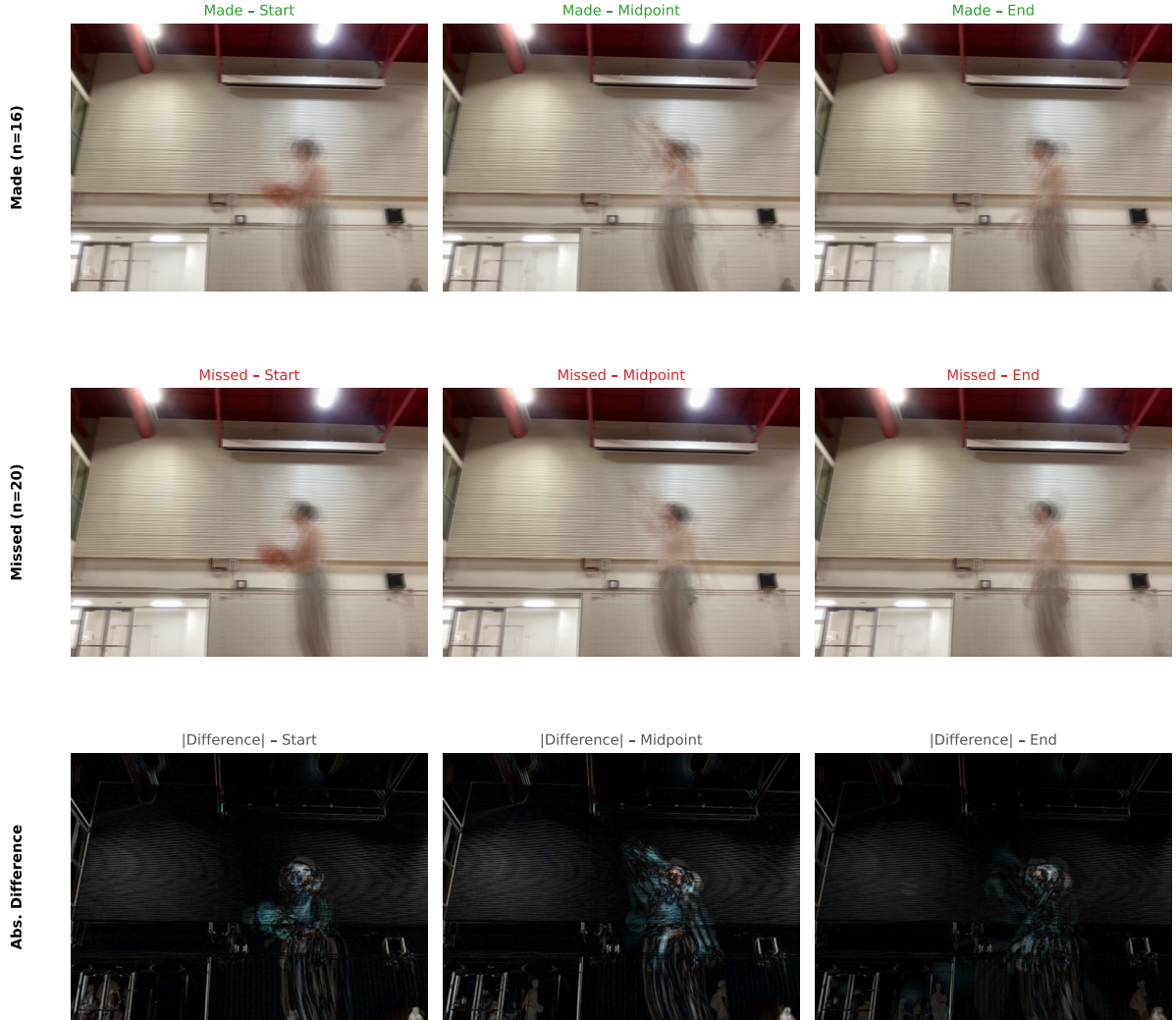


Figure 13: Average frame composites. Top row: made shots. Middle row: missed shots. Bottom row: absolute pixel difference. Columns correspond to execution start, midpoint, and end.

5 Discussion

This session (78 analysed shots across 3 blocks, 44% accuracy) demonstrates that the FreethrowEEG system can capture frequency-band-specific EEG dynamics during free-throw shooting using a portable Muse 2 headset with real EEG data.

Effect sizes for pre-shot band power differences were generally small. With 78 shots across 3 blocks, statistical power remains limited, though substantially improved over the initial pilot.

The θ/α ratio was higher for made shots, which is broadly consistent with prior literature linking theta/alpha dynamics to attentional focus during precision tasks. Replication with more trials is essential.

The shot-by-shot progression analysis provides a preliminary look at temporal dynamics across the session. Changes in band power over successive shots may reflect fatigue, adaptation,

or strategy adjustment, though the current data cannot distinguish between these.

Video-based pose estimation using MediaPipe revealed descriptive differences in shooting kinematics between made and missed shots. Stop-motion decomposition and ghost-overlay composites provide visual evidence of form consistency, while time-aligned pose and EEG traces offer a preliminary window into the brain-body dynamics underlying free-throw execution.

5.1 Limitations

This analysis is based on 78 shots across 3 blocks, after exclusion of trials with data quality issues and experimenter-flagged missed cues. Block 1's EEG recording terminated early, yielding only 3 usable shots from that block. Block 2 lacked video, limiting biomechanical analysis to Block 3 only. The Muse 2 provides only four channels, precluding source localization or detailed topographic analysis. Band power values represent averages across all channels, which may obscure hemisphere-specific effects. The participant noted increasing fatigue across blocks, particularly in Block 3, which may confound longitudinal comparisons. Future sessions with multiple participants and counterbalanced block ordering are needed to draw reliable conclusions.

6 Conclusion

This multi-block session demonstrates that the FreethrowEEG system can successfully record and analyse EEG band power during free-throw shooting with a portable Muse headset under real experimental conditions. With 78 valid shots, this represents a substantial step up from the initial 10-shot pilot, providing more robust estimates of band-power differences between made and missed shots. Exploratory analyses identified Theta band power as potentially differentiating made from missed shots. The addition of video-based pose estimation provides a complementary kinematic dimension, enabling time-aligned analysis of brain and body dynamics during shot execution. Ghost overlays and average frame composites offer intuitive visual summaries of shooting form variability. These results provide a foundation for larger-scale data collection with multiple participants and more rigorous hypothesis testing.